Data Cleaning & Preparation: Change Log

DecodeLabs Data Analytics | Project 1 | Batch 2026

1. Dataset Overview

Source file	Dataset for Data Analytics.xlsx (Sheet1)
Size	1,200 rows x 14 columns
Columns	OrderID, Date, CustomerID, Product, Quantity, UnitPrice, ShippingAddress, PaymentMethod, OrderStatus, TrackingNumber, ItemsInCart, CouponCode, ReferralSource, TotalPrice
Date range	2023-01-01 to 2025-06-30
Tool	Python (pandas) in Jupyter Notebook; fully reproducible script
Cleaned output	data/cleaned_data.csv (raw file left untouched)

2. Change Log

Change ID	Description	Impact	Status
CR001	Filled blank **CouponCode** values with the label "No Coupon". A blank means no coupon was used, so mode imputation would have wrongly assigned a coupon to those orders.	309 records preserved (blank in 25.8% of rows). No rows deleted.	Resolved
CR002	Confirmed **Date** is a valid datetime and exported in ISO 8601 format (YYYY-MM-DD).	0 invalid dates found; 0 incorrectly formatted dates in the exported file.	Resolved
CR003	Audited **OrderID** for duplicates (by ID and by full-row match).	0 duplicate IDs, 0 duplicate rows found. All 1,200 records retained.	Resolved (no action needed)
CR004	Trimmed whitespace and checked categorical columns (Product, PaymentMethod, OrderStatus, ReferralSource, CouponCode) for spelling or case variants.	0 values needed trimming; no variants found.	Resolved (no action needed)
CR005	Enforced 2-decimal precision on **UnitPrice** and **TotalPrice**; checked that TotalPrice = Quantity x UnitPrice.	0 mismatches; no values changed.	Resolved (no action needed)

3. Before vs. After

Metric	Before	After
Total rows	1,200	1,200
Missing values (all columns)	309	0
Blank CouponCode	309	0
"No Coupon" label	0	309

Duplicate OrderIDs	0	0
Incorrectly formatted dates	0	0

4. Verification Gate (Threshold for Project 2)

Checks were run on the exported file (*cleaned_data.csv*), reloaded with every column read as text, so the result reflects the actual deliverable.

Check	Requirement	Result
Unique identifiers	0 duplicate OrderIDs	**PASS** (0 duplicates)
Date formats	0 dates outside YYYY-MM-DD	**PASS** (0 invalid)
Completeness	0 missing values	**PASS** (0 missing)
Arithmetic integrity	TotalPrice = Quantity x UnitPrice	**PASS** (0 mismatches)

5. Notes & Assumptions

- The raw dataset was already largely clean; most steps were audits that found no issues, and they are logged so reviewers can see they were checked.
- Repeated CustomerIDs (11 repeats) were intentionally kept: they represent repeat customers, not duplicate records.
- "No Coupon" is an assumption that blank means no coupon was applied. This should be confirmed with the data owner if possible.
- The original file in the data folder was not modified; all changes were applied to a copy.